The Effect of Different Charging Strategies on EV Load Frequency Control

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***Abstract*—Due to the quick power reaction characteristic, the electrical vehicles (EVs) are used for load frequency control. An EV frequency control strategy is proposed considering the charging and discharging characteristics of the EV battery. Different from battery energy storage system, the availability of EVs largely depends on the charging behaviors. With different charging power and different charging time, three charging strategies are studied (dumb charging, smart charging and constant charging). The effects of EV load frequency control with three charging strategies are compared under a severe frequency event. The simulation results show that the EV load frequency control effect is closely related with the charging strategy. Considering the system load curve along a whole day, "smart charging" is the most suitable for preventing severe frequency event.**

Keywords-frequency control; electrical vehicle; charging strategy; severe frequency event

I. INTRODUCTION

With the increasing concerns over the environmental and energy problems, the concept of "Low-Carbon Economy" has become more and more popular [1-3]. Due to the de-carbon capability for transportation systems, EVs have attracted attentions around the world in recent years. The quick power response capability of EVs is especially suitable for frequency response of the power systems [4], [5]. Related researches have been carried out to study the method of utilizing EV loads as a demand response resource to serve the frequency regulation of the power system. Ota developed a frequency drop control which allows the EV to adjust its charging/discharging power with the frequency deviation [6], [7]. Liu used the EV clusters to provide dynamic frequency response to the power system in consideration of charging demand of EV users [8].

Due to the mobility of EV loads, the load frequency control (LFC) effect is largely affected by the EV charging strategies. Different charging strategies have been carried out in former studies. Based on different charging periods, Qian used "Smart Charging" and "Dumb Charging" in the modeling of EV clusters [9]. With lower charging power, Meng developed a "Constant Charging" with which EVs keep charging during the whole parking period [10].

In this paper, based on an EV frequency control strategy, different charging strategies are compared in case of an generation loss. The effect of different charging strategies on EV load frequency control is studied. In section II, an EV frequency control strategy is proposed in consideration of the charging characteristic and discharging characteristic of the EV battery. In section III, the travelling time of EV users is studied and three charging strategies and their charging curves along a whole day are compared. In section IV, the effect of different charging strategies on EV load frequency control is tested.

II. EV FREQUENCY CONTROL STRATEGY

A. EV Frequency Control Strategy

The quick power reaction characteristic makes EVs suitable for providing frequency response in case of a severe frequency event.

The primary target of EV charging is satisfying the users' charging demand. Consequently, the State of Charge (*SoC*) should be high enough when the EV plug out. When active power imbalance occur in the power system, the EVs change their charging status in sequence according to their real-time *SoC* as shown in Fig. 1.

The upper limit and lower limit of the battery *SoC* ($SoC_{\max}$ and $SoC_{\min}$) are set to avoid over-charge and over-discharge considering EV battery maintenance. The charging/discharging power of EV loads is adjusted by resetting $SoC_{\max}$.

In the V1G mode, EVs' charging power is adjusted according to the system frequency signal (Δf) through resetting the upper limit ($SoC_{\max}^{\mathrm{V1G}}$) as shown in Fig. 1 (b) and (c). When the frequency increases, the charging power increases. When the frequency decreases, the charging power decreases, the $SoC_{\max}^{\mathrm{V1G}}$ is calculated as shown in (1).

$$\begin{aligned} & SoC_{\max}^{\mathrm{V1G}} = SoC_{\max}^{\mathrm{normal}} + k \times \Delta f \\ & \Delta f = f - f_0 \\ & SoC_{\min} \le SoC_{\max}^{\mathrm{V1G}} \le 1 \\ & \Delta f \ge \Delta f_{\mathrm{cr}} \end{aligned} \tag{1}$$

where $SoC_{\max}^{\mathrm{normal}}$ is nominal high *SoC* set point of the EV battery; $SoC_{\max}^{\mathrm{V1G}}$ is the modified high *SoC* set point under V1G mode; $SoC_{\min}$ is the minimum *SoC* value for satisfying the travelling requirement of EV owners; f_0 is the system nominal frequency; f is the actual system frequency; Δf_{cr} is the value of frequency decrease when $SoC_{\max}^{\mathrm{V1G}} = SoC_{\min}$.

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If the frequency signal falls lower than Δf_{cr}, EVs turn into V2G mode to prevent further frequency drop. The discharging power of EVs is dynamically adjusted according to the Δf as depicted in Fig. 1 (d). The upper limit (SoC_{max}^{V2G}) is calculated as shown in (2).

$$\begin{aligned} SoC_{max}^{V2G} &= 1 + k \times \Delta f + (SoC_{max}^{normal} - SoC_{min}) \\ SoC_{min} &\le SoC_{max}^{V2G} \le 1 \\ \Delta f &\le \Delta f_{cr} \end{aligned} \tag{2}$$

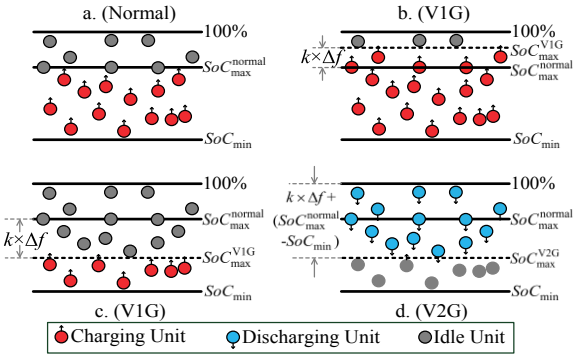


Figure 1. Diagram of EV frequency control strategy.

B. *Flow Chart of EV Frequency Control Strategy*

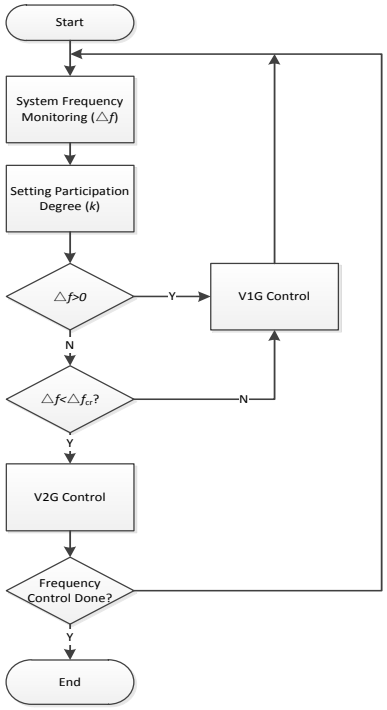


Figure 2. Flow chart of EV frequency control strategy.

III. EV CHARGING STRATEGIES

Due to the mobility of EV loads, their response capacity is dynamic rather than fixed. The travelling time of EVs depends on their type of use. HBW (Home-based-work) refers to the EVs owned by working individuals. HBO (Home-based-other) refers to the EVs owned by other individuals. NHB (Non-home-based) refers to the EVs owned by companies and organizations.

EVs are dynamically divided into a parking group and a non-parking group. The parking group is connected to the grid which is available for system frequency response. The controllable (parking) group is described as in (3).

$$t_{par} \le t \le t_{dep} \tag{3}$$

where t_{par} is the parking time of EV; t_{dep} is the departure time of EV.

The uncontrollable (non-parking) group is described as

$$t \notin [t_{par}, t_{dep}] \tag{4}$$

According to the transport behavior of EV owners, the ratio of the parking group during the whole day is shown in Fig. 3 [11, 12].

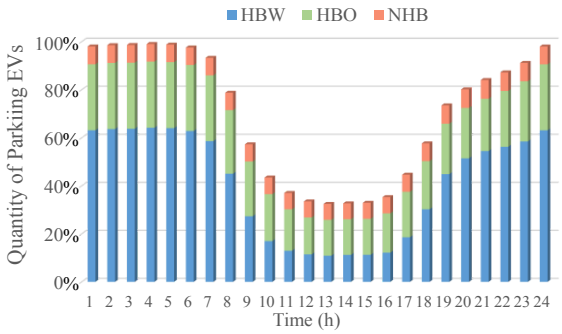


Figure 3. Quantity of parking EVs along a whole day.

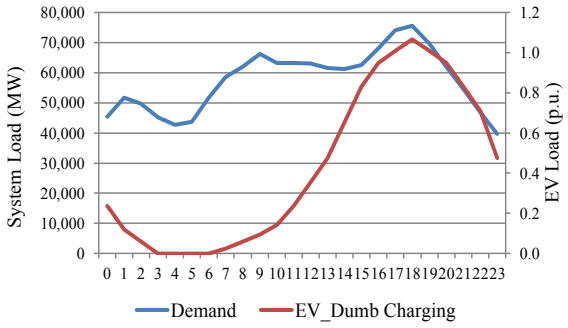


Figure 4. Load curve (Dumb Charging).

A. *Dumb Charging*

Dumb charging refers to traditional charging which means EVs start charging with rated charging power as soon as they are connected to the power grid. When the charging process is over ($SoC = SoC_{max}$), the EV plugs out and remain in idle status until departure. EVs are assumed to start charging at t_{par}. The probability distribution function of t_{par} is shown by (5).

$$f(t_{par}; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(t_{par}-\mu)^2/2\sigma^2} \tag{5}$$

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where $\mu = 18:00, \sigma = 4\text{h}$.

In this charging strategy, the peak load of EVs happen at 18:00~19:00, during which the system peak load usually occur. With large quantities of EVs connected to the power system, the system peak load will be further raised. The system load curve of a certain area and the EV load curve is shown as in Fig. 4.

B. *Smart Charging*

In most cases, the charging time of the EV is much longer than parking time especially during night period. In the case of time of use (TOU) price, the charging process can be delayed in order to lower the power consumption cost [13, 14]. The probability distribution function of t_{par} is shown by (6) where $\mu = 1:00, \sigma = 5\text{h}$. The system load curve of a certain area and the EV load curve is shown as in Fig. 5.

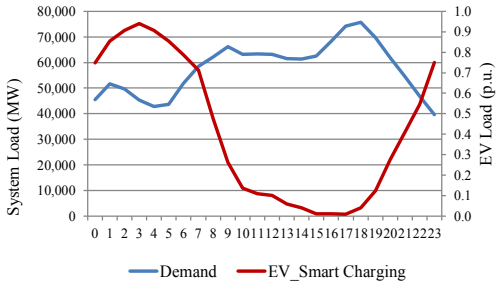


Figure 5. Load curve (Smart Charging).

In smart charging, on the one hand, the charging cost can be lowered under TOU. On the other hand, the EV peak load is transtered to 1:00 to avoid system peak load.

C. *Constant Charging*

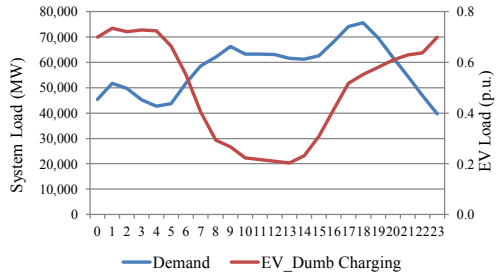


Figure 6. Load curve (Constant Charging).

V2G technology enables EVs to discharge to the power system in case of power vacancy in the power grid. To increase the discharging capacity a "Constant Charging" is proposed [10]. EVs are set to charge with a constant charging power during the whole parking period. In constant charging, the charging power P_0 is calculated as in (6).

$$P_0 = \frac{SoC_{\max} - SoC_{\text{start}}}{t_{\text{dep}} - t_{\text{par}}} \tag{6}$$

where SoC_{start} is the SoC at t_{par}.

All EVs connected to the power grid are in charging status. Consequently, the EV charging curve is closely related with the number of parking EVs. With constant charging, the system load curve of a certain area and the EV load curve is shown as in Fig. 6.

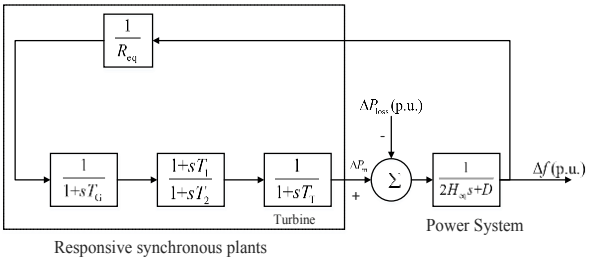


Figure 7. A simplified frequency regulation model.

IV. Case Study

In the case study, a simplified frequency regulation model based on SIMULINK/MATLAB platform is used to study the effect of different EV charging strategies on the EV LFC as in Fig. 7. $\triangle P_{\text{loss}}$ is used to represent the generation outage which leads to frequency drop in the power system.

The frequency characteristic of the power system after a generation loss depends on the system load and the scale of generation loss. Firstly, the system frequency characteristic is studied under different scales of generation loss. Secondly, the EV frequency control effect is tested under different EV quantities. Then, the frequency drops under different system loads are compared. Finally, the effect of the charging strategies on the EV LFC is tested.

Without EV LFC, the system frequency under different scales of generation loss is shown in Fig. 8 (system load= 50000 MW). Under a generation loss of 600 MW, the minimum frequency is 49.5Hz. Under a generation loss of 1200 MW, the minimum frequency is 49.1 Hz. Under a generation loss of 1800MW, the minimum frequency is 48.7Hz.

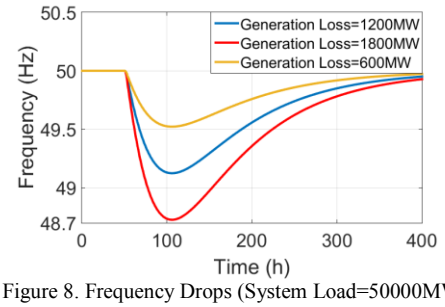


Figure 8. Frequency Drops (System Load=50000MW).

Without EV LFC, the system frequency after a generation loss under different scales of system load is shown in Fig. 9 (Generation Loss=1200MW). When the system load=50000MW, the minimum frequency is 49.1Hz. When the system load=55000MW, the minimum frequency is 49.2Hz. When the system load=60000MW, the minimum frequency is 49.3Hz.

With EV LFC, the system frequency with different quantities of EVs after a generation loss is shown in Fig. 10 (Generation Loss=1200MW, System Load=50000MW). When EV Load=0MW, the minimum frequency is 49.1Hz.

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When EV Load=200MW, the minimum frequency is 49.3Hz. When EV Load=400MW, the minimum frequency is 49.4Hz.

Fig. 11 shows the minimum frequency after the power system suffers a loss of 1200 MW generation at every half an hour with EV dumb charging. The EV LFC is able to largely promote the minimum frequency after a generation loss during 10:00~24:00. While during 0:00~10:00 the EV frequency control effect is not obvious due to the lack of charging EVs. Moreover, during 0:00~6:00 the minimum frequency is lower than 49Hz which is not allowed in the system operation.

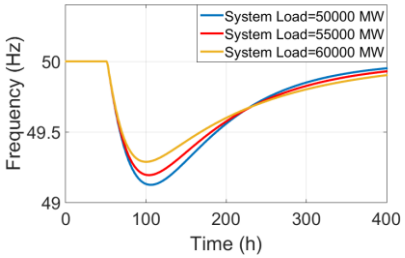


Figure 9. Frequency Drops (Generation Loss=1200MW).

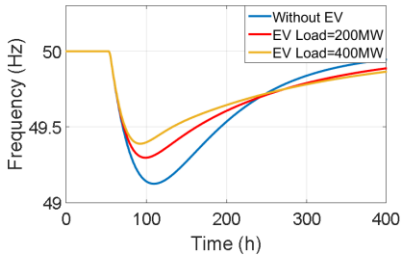


Figure 10. Frequency Drops (Generation Loss=1200MW, System Load=50000MW).

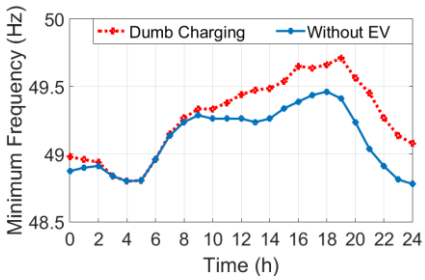


Figure 11. Minimum frequency after a generation loss every half an hour (Dumb Charging).

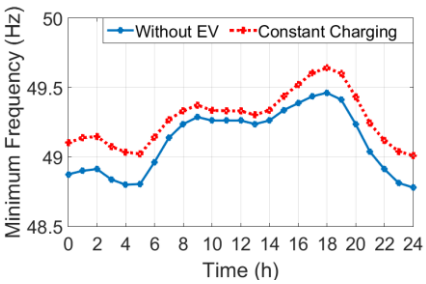


Figure 12. Minimum frequency after a generation loss every half an hour (constant charging).

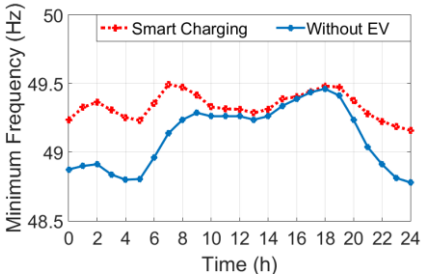


Figure 13. Minimum frequency after a generation loss every half an hour (Smart Charging).

Fig. 12 shows the minimum frequency after the power system suffers a loss of 1200 MW generation at every half an hour with EV constant charging. The EV LFC is able to promote the minimum frequency after a generation loss during most of the time along a whole day. However, the minimum frequency largely depends on system load. During valley load time (21:00~6:00) the system frequency nearly reaches 49Hz after the generation loss which will raise the risk of the system operation.

Fig. 13 shows the minimum frequency after the power system suffers a loss of 1200 MW generation at every half an hour with EV smart charging. The EV LFC effect largely depends on the quantity of charging EVs. During 6:00~21:00 the EV frequency control effect is not obvious due to the lack of charging EVs. While because of the high system load, the minimum frequency keeps in a high level during this period. During 21:00~6:00 the EV charging power is relatively high. Consequently, the minimum frequency can be promoted to an acceptable level after a generation loss during this period.

V. CONCLUSIONS

In this paper, EVs are used as demand response resource to provide frequency response service to the power system. An EV frequency control strategy is proposed considering both the charging characteristic and discharging characteristic. Then, three charging strategies are analyzed (dumb charging, constant charging, smart charging). The following conclusions can be drawn:

- EVs can provide effective frequency response to the power system. With EV LFC, the system frequency can be promoted after a generation loss.
- The EV LFC effect is affected by the system load and the quantity of EVs. Large quantities of EVs bring obvious frequency control effect.
- Considering the system load curve along a whole day, the “Smart Charging” is the most suitable for preventing severe frequency event after a generation loss.

Besides, other types of charging strategies can be used for other purposes such as providing dynamic frequency service. Different charging strategies can be used together as “Hybrid Charging” for demand response which needs further research in the future.

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